

Reliability and Safety

CompTIA SecAI+: AI Security Certification Complete Exam Prep 2026 | Section 24: Responsible AI

1. The Stakes of AI Reliability

Reliability and safety in AI are not abstract engineering concepts—they have direct human consequences when they fail. Reliability means an AI system performs its intended function correctly and consistently. Safety means an AI system does not cause unacceptable harm when it fails or encounters unexpected inputs. These are related but distinct requirements: a system can be reliable (consistently producing the same output) yet unsafe (if that output is consistently wrong in dangerous ways). Both require deliberate engineering across the full AI lifecycle, not just at the testing phase.

The NIST AI RMF 1.0 MAP and MEASURE functions are specifically designed to identify AI failure modes and evaluate performance before and after deployment. NIST CSF 2.0 maps reliability and safety obligations across multiple functions: ID.RA (Risk Assessment) identifies failure modes before deployment; PR.PS (Platform Security) maintains operational correctness; DE.AE (Adverse Event Analysis) detects reliability degradation; RS.AN (Incident Analysis) investigates failures; and RC.RP (Recovery Planning) restores safe operation afterward. Reliability is a cross-functional discipline—it does not live only in the testing phase.

2. Real-World Incident: Uber Autonomous Vehicle Fatality

Real-World Incident — Uber ATG / Elaine Herzberg (Tempe, Arizona, 2018): On March 18, 2018, an autonomous Uber vehicle struck and killed Elaine Herzberg—the first pedestrian fatality caused by an autonomous vehicle. The NTSB investigation found that the system detected Herzberg 5.6 seconds before impact but misclassified her repeatedly as an unknown object, then a vehicle, then a bicycle. The system had emergency braking disabled, and the human safety driver was watching a video on a personal device. Multiple compounding reliability failures—classification error, disabled safety override, and failed human oversight—produced a fatal outcome. This case is foundational for understanding how individual reliability failures cascade through a system.

The Uber case illustrates three key reliability concepts: distribution shift failure (the system encountered an edge case its training had not adequately represented), disabled fail-safe design (emergency braking was turned off to prevent erratic behavior during testing), and failed human oversight (the safety driver was not attending to the road). These are not independent failures—they are system design choices that compounded. Reliability engineering must account for how failure modes interact, not just evaluate each component in isolation.

3. Categories of AI Failure Modes

Failure Category	Definition	Primary Countermeasure
Distribution shift	Model encounters inputs materially different from training distribution	Production monitoring; automated retraining triggers

Adversarial failure	Crafted inputs cause intentional misprediction	Adversarial robustness testing; input validation
Cascading failure	One component failure propagates and amplifies through system stages	System-level FMEA; redundancy design
Performance failure	AI is technically correct but too slow for operational requirements	Latency SLAs; load testing
Integration failure	AI operates correctly in isolation but fails in connected system context	Integration testing; contract testing
Graceful degradation failure	System fails catastrophically instead of entering a defined safe state	Explicit safe state design; fallback mechanisms

Systematic reliability engineering enumerates failure modes through Failure Mode and Effects Analysis (FMEA) before they manifest in production. FMEA asks, for each component: what can fail, what is the effect, what is the severity, and what is the detectability? Applied to AI systems, FMEA extends to training data dependencies, inference infrastructure, model update pipelines, and downstream systems that consume AI outputs.

4. Testing for Reliability: Beyond Standard Accuracy

Standard ML evaluation measures accuracy on a held-out test set from the same distribution as training data. Reliability engineering requires going significantly further:

- Robustness testing — evaluates performance under distribution shift by testing on data from different time periods, geographies, or collection conditions than training data.
- Adversarial robustness testing — uses tools like Foolbox or IBM's open-source Adversarial Robustness Toolbox (ART) to quantify vulnerability to crafted adversarial perturbations; covered in the MITRE ATLAS v2.1 framework.
- Stress testing — evaluates whether accuracy holds under peak operational load; a model that degrades at 10x normal request volume has a reliability failure mode that may only surface during incidents.
- Integration testing — validates AI components in their full system context, not in isolation; tests that the AI's outputs are correctly consumed and acted on by surrounding systems.
- Regression testing — ensures model updates do not degrade performance on previously passing test cases; automated regression suites are essential for maintaining reliability through continuous development.

Automated testing pipelines incorporating all these test types are essential for maintaining reliability through continuous development cycles. A team that only runs accuracy evaluation before release is not doing reliability testing—it is doing baseline functional testing.

5. Safety Testing: The High-Stakes Layer

Safety testing answers a different question than reliability testing: not 'does this work correctly?' but 'when this fails, is the outcome acceptable?' This distinction is critical. A system can be highly reliable—failing rarely—yet cause catastrophic harm when it does fail. Safety testing methods include:

- Hazard analysis — identifies conditions under which AI failures cause harm; borrowed from safety-critical engineering in aviation and nuclear power, adapted for AI-specific failure modes.
- FMEA (Failure Mode and Effects Analysis) — systematically evaluates each component failure and its downstream consequences, with severity and detectability ratings.
- Safety case documentation — constructs an argument, with supporting evidence, that a system is sufficiently safe for its intended deployment context; required in automotive (ISO 26262) and medical device (IEC 62304) contexts.
- Red team exercises — simulate adversarial conditions and unexpected inputs to find failure modes before deployment; broader than adversarial ML testing, covering operational scenarios.
- Simulation testing — validates safety in controlled environments before real-world exposure; standard in healthcare, defense, and automotive AI development.

Safety testing is expensive relative to standard ML evaluation. Not doing it in safety-critical AI is more expensive—in human cost, legal liability, and organizational consequences.

6. Human Oversight Patterns

Pattern	Definition	Appropriate Context	EU AI Act Requirement?
Human-in-the-loop	Human approval required before any AI decision is acted on	High-consequence decisions; low AI reliability in context	Yes — for high-risk AI
Human-on-the-loop	Human monitors AI decisions and can intervene	Moderate-consequence decisions; high AI reliability	Situational
Human-in-command	Human sets goals and constraints; AI executes within bounds	AI augments human strategy; executes tactics autonomously	Situational

The EU AI Act mandates human oversight for all high-risk AI applications, making human-in-the-loop the legal default in regulated contexts including hiring, credit, healthcare, and law enforcement. Full automation is appropriate only where error consequences are low, reliability in the specific context is demonstrably high, and human review would introduce its own errors (e.g., slower response than the system can tolerate). The right pattern is a function of consequence severity and demonstrated reliability, not organizational preference.

7. Redundancy and Fail-Safe Design Principles

Redundancy is a foundational principle of reliable systems engineering, applied across multiple layers in AI deployments:

- Ensemble models — combining predictions from multiple models trained independently reduces the impact of any single model's failure modes; correlated failures are a risk if all models share the same training data.
- Diverse implementations — models from different vendors or architectures prevent correlated failures where all models fail simultaneously on the same unusual input.

- Fallback to simpler models — if a complex neural network fails or produces low-confidence outputs, fall back to a rule-based system that provides a reliable performance floor.
- Hardware redundancy — multiple inference servers behind a load balancer prevents infrastructure failure from causing AI system outages.

Memory anchor — Single Points of Failure: A single point of failure in a safety-critical AI system is an engineering defect, not an acceptable cost optimization. Redundancy has real cost; that cost is justified by the safety margin it provides. Any architecture diagram for safety-critical AI that shows a single model or single inference server without fallback should be treated as incomplete.

8. Production Monitoring: Detecting Failures Before Users Do

Deployed AI systems require continuous monitoring to detect reliability degradation before it causes harm. Four monitoring layers work together:

- Performance monitoring — tracks prediction latency, throughput, and error rates; detects infrastructure reliability failures.
- Accuracy monitoring — compares AI predictions against ground truth labels as they become available; detects accuracy degradation over time.
- Data drift monitoring — detects shifts in input feature distributions that predict accuracy degradation before it fully manifests; catches problems in advance rather than after users report errors.
- Anomaly detection in predictions — identifies unusual output patterns (clusters of extreme confidence scores, sudden changes in output distribution) that indicate something has changed in the model's operating environment.

Tools for production ML monitoring include Evidently AI (open-source Python), Arize AI, and Fiddler AI. Without production monitoring, reliability failures are discovered by users rather than detected by operators—a reactive posture that is characteristic of immature AI operations.

9. Distribution Shift: The Silent Reliability Killer

Distribution shift—when production data differs materially from training data—is the most common cause of AI reliability degradation in practice. Three types occur in production deployments. Covariate shift: the distribution of input features changes while the underlying relationship between features and outcomes remains the same. Concept drift: the relationship between inputs and outputs changes, so the model's learned function no longer reflects reality. Dataset shift: both the input distribution and the input-output relationship change simultaneously.

A lending model trained on pre-2020 economic data experienced severe covariate shift when COVID-19 dramatically altered employment patterns, income volatility, and credit behavior. Models that were highly accurate before March 2020 became unreliable within weeks—faster than standard retraining cycles could respond. Detecting distribution shift requires monitoring input feature statistics against training distribution baselines. Automated retraining triggers when drift exceeds defined thresholds can partially address this, but human review of retrained models is essential before automatic redeployment.

10. Reliability as an Attack Surface

Reliability failures are exploitable attack surfaces, not just engineering quality concerns. Two attack patterns deserve specific attention. First, adversarial examples crafted to cause misprediction—documented in MITRE ATLAS v2.1 as a named attack technique. An attacker who understands a model's training distribution can craft inputs that appear legitimate but systematically cause the model to fail in targeted ways. Second, reliability attacks under load: a targeted denial-of-service attack against an AI inference layer can degrade response time beyond operational thresholds, effectively disabling AI-dependent security controls. A fraud detection system that cannot respond within its SLA during a DDoS event is as useless as one that has been compromised.

Security practitioners should test AI reliability under adversarial load conditions, not just under normal operation. The combination of adversarial input testing and load testing under attack conditions provides a more complete reliability picture than either test in isolation.

11. Graceful Degradation: Designing for Failure

Perfect reliability is unachievable in any complex system. Graceful degradation designs AI systems to fail safely rather than catastrophically. Four design principles enable graceful degradation:

- Reduced functionality — if the AI component fails, the system continues operating with reduced capability rather than going offline entirely; users experience degraded service rather than service outage.
- Conservative defaults — when the AI is uncertain, it makes the safer choice rather than guessing; a diagnostic AI that cannot reach a confident conclusion flags for human review rather than outputting a low-confidence diagnosis.
- Clear communication — when operating in degraded mode, the system communicates its limitations transparently rather than presenting degraded outputs as full-capability results.
- Safe state transitions — when a failure is detected, the system automatically transitions to a defined safe state; automotive AI that loses critical sensor data transfers control to the driver rather than attempting to operate blind.

Graceful degradation requires explicit design decisions. It does not emerge from fixing individual bugs. A team that has not explicitly designed and tested safe state transitions for their AI system has not implemented graceful degradation—they have assumed it.

12. Verification vs. Validation: The Critical Distinction

Two distinct quality disciplines govern AI system correctness, and confusing them is a common source of production failures. Verification asks: does the system do what the specification says? It measures technical correctness—does the model achieve specified accuracy on the validation set, does inference latency meet the specified SLA, do all unit tests pass? Validation asks: does the specification describe the right thing? It measures fitness for purpose—does validated accuracy correspond to safe performance in the real deployment context, are we measuring what actually matters, is the population we tested on representative of who will actually use the system?

The distinction matters because a system can pass all verification tests while failing validation—correctly implementing a flawed specification. A medical AI that meets its technical accuracy specification but is deployed on a patient population outside its validation scope has passed verification and failed validation. Both disciplines are necessary, and validation often requires domain expertise that goes beyond the AI engineering team. Involving clinical staff, domain specialists, and affected community representatives in validation is a Responsible AI requirement, not optional consultation.

13. Safety Standards by Domain

Domain	Standard(s)	Scope
Automotive	ISO 26262, ISO 21448 (SOTIF)	Functional safety and safety of the intended functionality for road vehicles
Aviation	DO-178C	Software considerations in airborne systems and equipment certification
Medical devices	FDA AI/ML-based SaMD framework, IEC 62304	AI/ML software as a medical device; software lifecycle requirements
General AI governance	NIST AI RMF 1.0	Voluntary framework for AI risk management across sectors
International AI management	ISO/IEC 42001:2023	Certifiable AI governance management system standard

These domain-specific standards share a common structure: documented risk management, systematic testing evidence, traceability from requirements to tests, and formal approval processes. Their overhead is justified by the reduction in catastrophic failure risk. Commercial aviation's extraordinary safety record demonstrates what systematic safety engineering achieves at scale.

Key Terms Glossary

Term	Definition
Reliability (AI)	AI system performs its intended function correctly and consistently across its operating range
Safety (AI)	AI system does not cause unacceptable harm when it fails or encounters unexpected inputs
Distribution shift	When production input data differs materially from training data, causing reliability degradation
Covariate shift	The distribution of input features changes while the relationship between features and outcomes remains constant
Concept drift	The relationship between inputs and outputs changes, so the model's learned function no longer reflects reality
FMEA	Failure Mode and Effects Analysis; systematic technique enumerating component failures and their downstream consequences
Human-in-the-loop	Oversight pattern requiring human approval before any AI decision

	is acted on; mandated for EU AI Act high-risk AI
Graceful degradation	System design that produces reduced functionality rather than catastrophic failure when components fail
Verification	Confirming a system does what its specification says; measures technical correctness
Validation	Confirming the specification describes the right thing; measures fitness for deployment purpose
SOTIF (ISO 21448)	Safety of the Intended Functionality; automotive standard addressing AI safety for scenarios without component malfunction
Adversarial Robustness Toolbox	IBM open-source library for testing AI robustness against adversarial examples and evaluating defenses

Quick Revision Checklist

- State the distinction between reliability (correct function) and safety (acceptable failure behavior).
- Name the six AI failure mode categories: distribution shift, adversarial, cascading, performance, integration, graceful degradation.
- Explain how distribution shift, disabled fail-safe design, and failed human oversight combined in the Uber Tempe fatality.
- Distinguish verification (does it meet the spec?) from validation (is the spec right?).
- Know the three human oversight patterns: human-in-the-loop, human-on-the-loop, human-in-command — and which is mandated for EU AI Act high-risk AI.
- Name three reliability testing types beyond accuracy: robustness, adversarial robustness, stress testing.
- Explain three distribution shift types: covariate shift, concept drift, dataset shift.
- Describe four graceful degradation design principles: reduced functionality, conservative defaults, clear communication, safe state transitions.
- Name three production monitoring layers: performance monitoring, accuracy monitoring, data drift monitoring.
- Know domain safety standards: ISO 26262/21448 (automotive), DO-178C (aviation), FDA AI/ML SaMD framework and IEC 62304 (medical devices).

10 Exam-Based MCQs — Reliability and Safety

Test yourself after reading. These questions are written in real exam style — scenario-heavy, with plausible distractors. Read every explanation, including for questions you answered correctly.

Q1. *Scenario:* During a review of the 2018 Uber ATG fatality investigation, a security engineer is asked to categorize the AI failure modes that contributed to the outcome. The NTSB report documents that the autonomous system detected the pedestrian 5.6 seconds before impact, misclassified the object as unknown, then a vehicle, then a bicycle, and had emergency braking disabled. The human safety driver was not monitoring the road. An autonomous vehicle AI system that detects a pedestrian misclassifies

the object repeatedly before impact, and the emergency braking override has been disabled during testing. Which AI failure category MOST directly contributed to the outcome?

- A) Integration failure, because the AI and braking system were not tested together
- B) Distribution shift failure combined with graceful degradation failure
- C) Adversarial failure, because the pedestrian's movement pattern confused the classifier
- D) Performance failure, because the inference was too slow to respond in time

Answer: B **Explanation:** B is correct. The misclassification pattern—repeatedly cycling between object categories—is characteristic of distribution shift failure: the system encountered an edge case (a person crossing outside a crosswalk at night) that its training distribution had not adequately covered. The disabled emergency braking is a graceful degradation failure: the safe-state behavior (emergency stop) had been removed, so when the primary classification failed, there was no fallback. A is wrong; the components communicated but the braking was intentionally disabled. C is wrong; there is no indication of crafted adversarial input. D is wrong; the issue was classification error, not response latency.

Q2. Scenario: A risk analytics team reviews model performance reports for a retail lending platform deployed across three regional banks. The AI model produces default probability scores used in credit decisions. Performance reports show accuracy falling from 87% to 61% between March and June 2020, with no changes to the model, infrastructure, or input data pipeline. The data volume and feature schema remain identical to pre-deployment periods. A lending institution's AI model was trained on data from 2016 to 2019 and deployed in early 2020. By mid-2020, default prediction accuracy had fallen from 87% to 61%. The input data volume and format remained unchanged. What is the MOST likely cause?

- A) Adversarial attack by a competitor manipulating loan application data
- B) Representation bias because minority borrower applications increased during the period
- C) Concept drift caused by COVID-19 changing the relationship between features and default probability
- D) Integration failure because the model was connected to a new data pipeline in 2020

Answer: C **Explanation:** C is correct. Concept drift occurs when the relationship between inputs and outputs changes—the model's learned function no longer reflects reality. COVID-19 dramatically altered the economic meaning of employment status, income level, and credit utilization patterns: features that predicted credit behavior in 2016-2019 had different predictive relationships to default probability in mid-2020. A is wrong; there is no evidence of adversarial manipulation in the scenario. B is wrong; representation bias causes demographic performance disparities, not an overall 26-point accuracy drop. D is wrong; the scenario explicitly states the data pipeline was unchanged.

Q3. Which combination of oversight requirements does the EU AI Act mandate for AI systems used in employment decisions?

- A) Mandatory explainability using LIME or SHAP, plus annual fairness audits
- B) Human-in-the-loop approval, data governance controls, and technical transparency documentation
- C) Human-on-the-loop monitoring with automated fairness metric alerting

D) ISO 26262 functional safety certification before deployment

Answer: B **Explanation:** B is correct. The EU AI Act classifies AI in employment decisions as high-risk, requiring human oversight (human-in-the-loop for consequential decisions), strict data governance, and technical documentation including transparency requirements. A is wrong; the EU AI Act does not mandate specific explainability tools like LIME or SHAP, though explainability methods support compliance. C is wrong; human-on-the-loop is a weaker oversight pattern than the Act's high-risk requirements. D is wrong; ISO 26262 is an automotive functional safety standard, not applicable to HR AI.

Q4. What is the PRIMARY purpose of data drift monitoring in a production AI system, as distinct from accuracy monitoring?

- A) Data drift monitoring detects model accuracy degradation after it has occurred in production
- B) Data drift monitoring detects shifts in input distributions that predict accuracy degradation before it fully manifests
- C) Data drift monitoring identifies adversarial inputs designed to cause model misprediction
- D) Data drift monitoring tracks changes in the model's output distribution to detect bias emergence

Answer: B **Explanation:** B is correct. Data drift monitoring watches input feature distributions against training baselines. When input distributions shift significantly, accuracy degradation typically follows—monitoring the drift provides an early warning before users experience failures. This is the key distinction from accuracy monitoring, which detects degradation after it has occurred. A describes accuracy monitoring, not drift monitoring. C describes adversarial input detection, a different monitoring concern. D describes output distribution monitoring, which is a valid separate technique but not what data drift monitoring specifically does.

Q5. A medical AI diagnostic system meets all technical accuracy specifications in validation testing but is deployed to a patient population that differs significantly from the validation cohort in age distribution and comorbidity patterns. Which quality discipline failure does this represent?

- A) Verification failure — the system did not meet its specified accuracy on the validation set
- B) Validation failure — the specification does not describe the right thing for the actual deployment context
- C) Reliability failure — the model will produce distribution shift errors in production
- D) Safety failure — the system's fail-safe mechanisms were not tested on the deployment population

Answer: B **Explanation:** B is correct. Validation failure occurs when a system passes all technical tests but those tests do not reflect the actual deployment context. The system correctly implements its specification (passing verification) but the specification was written for the wrong population. The validation test set does not represent who will actually use the system, so passing it is insufficient evidence of deployment safety. A is wrong; the scenario explicitly states the system meets its accuracy specification. C is a consequence of the validation failure but does not name the quality discipline failure. D is a possible downstream concern but not what is directly described.

Q6. Scenario: During a DDoS attack targeting the authentication infrastructure of a regional bank, the security operations center observes that the AI-powered real-time fraud detection system is timing out on approximately 73% of transaction scoring requests. Transactions that would normally be flagged for review are being approved automatically due to a system design decision that defaults to approval when the AI scoring service is unavailable. An AI fraud detection system is experiencing degraded response times during a coordinated attack on the bank's web infrastructure. The AI inference layer is responding in 8 seconds instead of the SLA-mandated 400 milliseconds, effectively bypassing fraud checks for real-time transactions. Which concept does this MOST directly illustrate?

- A) Adversarial example attack exploiting model decision boundaries
- B) Reliability attack using load conditions to disable AI-dependent security controls
- C) Graceful degradation failure because the system lacks a lower-capability fallback mode
- D) Distribution shift causing the model to misclassify fraudulent transactions as legitimate

Answer: B **Explanation:** B is correct. This scenario describes a reliability attack using load conditions to disable AI security controls by exceeding their operational SLA thresholds. The fraud detection AI is not malfunctioning internally—it is being rendered operationally ineffective by an external attack on the infrastructure it depends on. A is wrong; adversarial examples target model prediction, not response time. C is a secondary concern (what should happen when the SLA is breached) but does not name the primary attack concept. D is wrong; the model's classification logic is not being affected—its availability is.

Q7. Which approach to AI redundancy MOST directly addresses the risk that multiple models might fail simultaneously on the same unusual input?

- A) Ensemble models combining predictions from multiple models trained on the same dataset
- B) Diverse implementations using models from different vendors or trained with different architectures
- C) Hardware redundancy using multiple inference servers behind a load balancer
- D) Fallback to a rule-based system when the primary neural network produces low-confidence outputs

Answer: B **Explanation:** B is correct. The risk of simultaneous failure on the same unusual input is called correlated failure. Ensemble models (A) can suffer correlated failures if all constituent models share the same training data and therefore the same blind spots. Diverse implementations using models from different vendors, trained on different data with different architectures, reduces correlated failure risk because different models have different failure modes. C addresses infrastructure failure, not model-level correlated failures. D provides a fallback but does not address the correlated failure risk in the primary layer.

Q8. A team has completed extensive unit testing and integration testing of an AI system. All tests pass. Which additional testing type is MOST critical to add before production deployment of a safety-critical AI?

- A) Regression testing to ensure model updates do not degrade previously passing cases

- B) Safety case documentation providing an argued, evidence-supported claim the system is sufficiently safe
- C) Stress testing at 10x normal load to verify accuracy holds under peak conditions
- D) Adversarial robustness testing using Foolbox to quantify vulnerability to crafted inputs

Answer: B **Explanation:** B is correct for safety-critical AI specifically. Safety case documentation is the formal mechanism by which safety engineers argue, with supporting evidence, that a system is sufficiently safe for its intended deployment context—it goes beyond test results to synthesize all safety evidence into a documented claim. This is required for medical devices (IEC 62304), automotive (ISO 26262), and aviation (DO-178C) contexts. A, C, and D are all valid and important testing types, but none specifically constitutes safety case documentation for a safety-critical context. The question asks what is MOST critical given that unit and integration testing are already complete.

Q9. What distinguishes human-in-the-loop from human-on-the-loop oversight in AI systems?

- A) Human-in-the-loop requires the human to understand the AI's internal model; human-on-the-loop does not
- B) Human-in-the-loop requires human approval before any decision is acted on; human-on-the-loop allows autonomous action with human ability to intervene
- C) Human-in-the-loop is mandatory under the EU AI Act for all AI; human-on-the-loop is optional
- D) Human-in-the-loop applies to automated systems; human-on-the-loop applies only to AI with decision support functions

Answer: B **Explanation:** B is correct. Human-in-the-loop requires human approval at a defined decision gate before any AI decision is acted upon—the AI cannot proceed without human authorization. Human-on-the-loop allows the AI to act autonomously while a human monitors and retains the ability to intervene and override. A is wrong; neither pattern requires understanding the model's internals. C is wrong; the EU AI Act mandates human oversight for high-risk AI specifically, not all AI, and the specific pattern required is human-in-the-loop for high-risk contexts. D is incorrect in its characterization of when each applies.

Q10. An AI system deployed in a hospital for triage prioritization fails to reach a confident classification for a patient presentation. What is the MOST appropriate graceful degradation response?

- A) Output the highest-confidence prediction available, even if below the system's standard threshold
- B) Flag the case for immediate human clinical review rather than assigning an automated priority
- C) Default the patient to the lowest priority queue to avoid false positive emergency escalations
- D) Retry the classification three times before escalating to human review

Answer: B **Explanation:** B is correct. The graceful degradation principle of conservative defaults requires that when an AI is uncertain, it makes the safer choice. In a medical triage context, the safe choice when uncertain is to route the case to a human clinician rather than assigning an automated priority that may be wrong. C is wrong because defaulting to lowest priority is not conservative—it risks undertreating a genuine emergency. A is wrong because outputting a below-threshold prediction violates the conservative

defaults principle and communicates false confidence. D introduces delay before escalating, which in triage could directly affect patient outcomes.